When Bigger Is Not Always Better: Why Model Scale Alone Fails at Fine-Grained Document Classification

Naa Lamiorkor Boye
MSc IT, Digitalisation and Sustainability
Lucerne University of Applied Sciences and Arts (HSLU)
Rotkreuz, Switzerland
naalamiorkor.boye@stud.hslu.ch
GitHub Repository: github.com/Lamiorkor/NALAPRO_Final_Project

***Abstract*—This paper studies how representation, supervision, and task adaptation affect topic classification on the 20 Newsgroups dataset. The same underlying problem was approached progressively: averaged Word2Vec embeddings and TF–IDF features were first supplied to a deliberately simple two-layer neural network; BERT was then fine-tuned directly for classification; BERT was additionally continued on the task corpus with masked language modeling before supervised fine-tuning; and Llama-3 was evaluated through zero-shot prompting, few-shot prompting, a full-context diagnostic, and QLoRA fine-tuning. The strongest directly comparable full-test model was the MLM-adapted BERT classifier, which achieved 74.01% accuracy and 0.7370 weighted F1 on 7,317 cleaned test documents. A TF–IDF unigram–bigram model reached 68.75% accuracy, showing that exact words and short phrases remained highly informative. Prompt-only Llama-3 was considerably weaker: zero-shot accuracy was 34.20%, while one-example-per-class few-shot prompting decreased it to 31.75%. QLoRA reached 75.00% accuracy and 0.7434 weighted F1 on a smaller 500-document subset, but its macro F1 was 0.6894, and the evaluation was not matched to BERT. The experiments, therefore, support a more careful conclusion than a simple model leaderboard. Better representations helped, but performance improved most when the learning objective matched the task. The project also changed my initial assumptions: dense semantic vectors did not automatically beat sparse lexical features, few-shot examples did not automatically help a large language model, and model scale alone did not compensate for weak task alignment.**

***Index Terms*—text classification, 20 Newsgroups, Word2Vec, TF–IDF, n-grams, BERT, masked language modeling, task-adaptive pretraining, Llama-3, QLoRA, prompt engineering**

I. Introduction

Natural language processing systems cannot work directly with meaning in the way people experience it. They work with representations: tokens, counts, vectors, contextual states, or generated text. The choice of representation determines what evidence is preserved, what is discarded, and what a classifier can learn. This becomes especially visible in topic classification. A sparse representation may preserve an exact phrase such as "gun control," while a dense vector may connect it to semantically related language. A contextual Transformer can interpret the phrase within a longer argument. A generative large language model can reason over a prompt, but it must still map its understanding to the exact labels required by the dataset.

This study was conducted as the final project for the Natural Language Processing (NALAPRO) module offered during the Spring Semester 2026 at the Lucerne University of Applied Sciences and Arts (HSLU). The project was designed to integrate and evaluate the major concepts covered throughout the course, including text preprocessing, TF–IDF, word embeddings, Transformer architectures, BERT fine-tuning, prompt engineering, and large language models.

This project investigates these differences using 20 Newsgroups, a benchmark of online discussion posts assigned to 20 topics [1]. The dataset is useful because some classes are easy to separate, while others share vocabulary and subject matter. For instance, the model must distinguish `rec.sport.baseball` from `rec.sport.hockey`, four closely related computer-system categories, three political categories, and several religion-related categories. As such, broad semantic recognition is not enough because the decision boundary itself matters.

The work began as a required comparison of Word2Vec and TF–IDF using the same simple neural network. I initially expected the learned dense representation to be clearly stronger because Word2Vec encodes semantic similarity. The result challenged that expectation. Word2Vec improved greatly as its embeddings trained for more epochs, yet TF–IDF was still stronger. Adding bigrams improved it further. That result motivated the central idea of this paper: a representation is not "better" in isolation; it is better when the information it retains fits the task.

The later experiments extend this idea. BERT contributes contextual, bidirectional representations and is optimized directly for the 20 labels. Continued masked-language-model (MLM) pretraining asks whether more exposure to the target corpus improves the encoder before classification. Llama-3 changes the setup more radically by turning classification into conditional generation. Finally, QLoRA tests whether lightweight task-specific training can close the gap between general instruction following and exact closed-set classification. By introducing a generative giant like Llama-3 into the mix, the study shifts from a simple benchmark evaluation to a fundamental investigation of whether massive model scale can compensate for a lack of explicit task alignment in fine-grained classification domains.

Beyond evaluating model performance, the project served as an opportunity to apply the progression of techniques introduced throughout the NALAPRO curriculum, moving from classical NLP representations to modern Transformer-based and generative language models. Consequently, the experimental design mirrors the conceptual development of the course itself.

The study is organized around three research questions:

1) **RQ1:** How does document representation affect a deliberately simple neural classifier when the architecture is held approximately constant?
2) **RQ2:** Does continued MLM pretraining on the unlabeled task corpus improve BERT after supervised fine-tuning?
3) **RQ3:** How do prompt-only and parameter-efficient Llama-3 approaches compare with models trained directly for the classification objective?

The paper contributes a linked experimental narrative rather than a collection of isolated scores. It reports representation-level evidence for Word2Vec, hyperparameter results for BERT, confusion-based error analysis for Llama-3, and a careful separation between full-test and subset evaluations. It also documents the methodological decisions and constraints that shaped the project, including preprocessing, truncation, compute access, experiment tracking, and the limits of visually comparing independently trained embedding spaces.

II. Background and Relation to Course Theory

A. The NLP Pipeline and Representation Choices

The theoretical foundations presented in this section reflect the sequence of concepts explored in the NALAPRO course, beginning with classical text representation techniques and progressing toward contextual language models and large language models. The course introduced NLP as a pipeline in which raw text is cleaned, tokenized, transformed, and supplied to an analytical model. Tokenization defines the units a system can process. Stopword removal reduces frequent terms that may carry little topic information. Stemming or lemmatization can collapse surface forms, while part-of-speech information can preserve grammatical roles. These choices are not neutral. Removing too much can erase useful meaning, while removing too little can allow noise or metadata to dominate.

This project deliberately uses two levels of preprocessing. Classical models receive lowercased, token-filtered text because their features depend directly on word identity and frequency. Transformer models receive less aggressively processed text because their pretrained tokenizers and contextual layers were learned from natural sequences. The distinction reflects an important lesson from the course: preprocessing should support the assumptions of the representation rather than being applied as a fixed ritual.

B. TF–IDF, N-Grams, and Explicit Lexical Evidence

A bag-of-words model represents a document by the presence or frequency of vocabulary items and ignores their order. TF–IDF improves on raw counts by reducing the influence of terms that appear throughout the corpus and increasing the influence of terms that are distinctive within a document [2], [3]. A common formulation is

$$\mathrm{tfidf}(t,d) = \mathrm{tf}(t,d)\log\left(\frac{N}{\mathrm{df}(t)}\right), \tag{1}$$

where t is a term, d is a document, N is the number of documents, and $\mathrm{df}(t)$ is the number of documents containing the term.

The benefit for 20 Newsgroups is intuitive. Words such as *hockey*, *encryption*, *motorcycle*, or *forsale* are not merely semantically meaningful; they are discriminative identifiers of classes. N-grams extend this representation by preserving short local sequences. They still do not model context in the Transformer sense, but a bigram can distinguish a phrase from its separate words. This is why the additional TF–IDF experiment used both unigrams and bigrams. It tested whether a simple classifier could improve without changing its architecture, only the evidence available to it.

C. Word2Vec and Distributional Learning

Word2Vec learns dense vectors from the distributional principle that words occurring in similar contexts tend to have related meanings [4], [5]. The skip-gram version predicts surrounding context words from a target word within a sliding window. Negative sampling makes this practical by training the model to distinguish observed target–context pairs from sampled non-neighbor pairs.

The learned vectors support cosine similarity:

$$\cos(\mathbf{x},\mathbf{y}) = \frac{\mathbf{x}^\top\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}. \tag{2}$$

Words used in similar contexts should receive similar directions in the embedding space. However, the project requires document rather than word classification. Each document was therefore represented by the arithmetic mean of its in-vocabulary word vectors:

$$\mathbf{v}_d = \frac{1}{|W_d|}\sum_{w\in W_d}\mathbf{v}_w. \tag{3}$$

This conversion is simple and fast, but it removes order, treats all retained words equally, and can dilute one highly diagnostic term among many generic terms. Arora *et al.* show that weighted averaging and removal of common directions can create stronger sentence representations [6]; the unweighted mean was retained here to keep the experiment aligned with the assignment and make the representation itself easy to interpret.

D. From Recurrent Models to Transformers

The course positioned Transformers within a longer development from recurrent neural networks (RNNs) and long short-term memory (LSTM) networks. RNNs process a sequence through recurrent hidden states, but long dependencies and sequential computation make them difficult to train and parallelize. LSTMs introduce a cell state and gated updates to retain or discard information over longer sequences [7], [8]. Their forget, input, and output gates address part of the vanishing-gradient problem, but the recurrent dependency remains.

Transformers replace recurrence with attention over the sequence [9]. For query, key, and value matrices Q, K, and V, scaled dot-product attention is

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V. \tag{4}$$

Self-attention allows every token to update its representation using other tokens directly, while positional encodings preserve order information. Multi-head attention lets the model learn several relationship patterns in parallel. This theoretical progression matters to the results: TF–IDF and averaged Word2Vec discard most order, whereas BERT can condition every token on the full visible sequence.

E. BERT, Pretraining, and Continued MLM

BERT is an encoder-only Transformer pretrained to form deep bidirectional representations [10]. Its MLM objective masks selected tokens and predicts them using context on both sides. Pretraining creates general language representations; supervised fine-tuning then adapts the network and a classification head to a specific label space. The public BERT release emphasized this pretrain-then-fine-tune paradigm and its ability to transfer one pretrained model across multiple downstream tasks [11].

Task 3 adds another stage between general pretraining and supervised classification. The base checkpoint is continued on the unlabeled 20 Newsgroups training text using MLM. Gururangan *et al.* distinguish adaptation to a broad domain from adaptation to the unlabeled corpus of a particular task [12]. Because the present model is continued on the same task distribution used later for classification, "task-adaptive pretraining" is the more precise term. The expected benefit is modest but meaningful: the model can become more familiar with forum-style text and technical vocabulary before learning the labels.

F. In-Context Learning, Scale, and Task Adaptation

Autoregressive language models can perform tasks from natural-language instructions and demonstrations without gradient updates. GPT-3 established zero-shot, one-shot, and few-shot evaluation as an important large-model paradigm [13]. Llama-3 follows the decoder-only generative family and can be prompted to emit a class label [14]. Yet a prompted classifier differs from a supervised encoder. It must interpret the instruction, infer the meaning of an artificial label inventory, generate a valid output, and choose among fine-grained categories. Each step introduces sensitivity.

Research has shown that few-shot performance depends on demonstration selection, ordering, formatting, and label calibration [15], [16]. Moreover, scale is not a universal guarantee of correct behavior. TruthfulQA, for example, was designed to test whether models reproduce common human falsehoods and found that larger models were not uniformly more truthful [17]. That benchmark is different from 20 Newsgroups, but the broader lesson is relevant: model size does not remove the need for an aligned objective and a careful evaluation design.

LoRA introduces trainable low-rank updates while keeping the base model frozen [18]. QLoRA combines this with a quantized base model to reduce memory use while retaining effective adaptation [19]. In this project, QLoRA tests whether Llama-3's low prompt-only accuracy is a limitation of model knowledge or a limitation of how that knowledge is connected to the 20 labels.

III. Dataset and Experimental Design

A. Dataset and Label Structure

The dataset was downloaded at runtime through scikit-learn's `fetch_20newsgroups` interface [20]. The original split contains 11,314 training documents and 7,532 test documents across 20 classes. The repository does not store the raw dataset. Table I summarizes the principal dataset sizes used throughout the study.

The labels form several natural families: five computer categories, four recreational categories, four science categories, three political categories, three religion-related categories, and one marketplace category. These families help interpret errors. Confusing `rec.sport.hockey` with `rec.sport.baseball` is different from confusing hockey with cryptography: the first indicates correct broad-domain recognition but weak fine-grained discrimination.

B. Cleaning and Sample Counts

Headers, footers, and quoted text were removed for the main cleaned setting. This decision reduces the risk of shortcut

TABLE I
Dataset statistics used in the experiments.

Statistic	Value
Original training documents	11,314
Original test documents	7,532
Number of classes	20
Cleaned non-empty training documents	11,014
Cleaned non-empty test documents	7,317
Token-cleaned training documents	11,000
Token-cleaned test documents	7,302
BERT training split	9,912
BERT validation split	1,102
Llama prompting subset	2,000
QLoRA training subset	2,000
QLoRA validation subset	200
QLoRA test subset	500

learning from e-mail addresses, subject-line patterns, signatures, and repeated thread material. It also makes the task harder in a useful way: models must rely more heavily on the content of the post.

For Word2Vec and TF–IDF, text was lowercased, numbers and punctuation were removed, NLTK tokenization was applied, alphabetic tokens were retained, and English stopwords were removed [21]. The Transformer experiments retained more natural text after the dataset-level removal because BERT and Llama-3 have pretrained tokenizers and contextual expectations.

The resulting sample counts are summarized in Table I. A reproducibility detail emerged during the project: "empty document" can mean two different things. Filtering raw cleaned strings produced more documents than filtering after token-level cleaning because some posts consisting only of punctuation, numbers, or stopwords became empty during later preprocessing stages. The report therefore prints evaluation size beside each result rather than presenting all numbers as if they came from an identical test matrix.

The BERT training portion used a 90/10 stratified split of the cleaned non-empty training data, resulting in 9,912 training documents and 1,102 validation documents. Prompt-based Llama evaluation used 100 examples per class, creating a balanced 2,000-document subset. QLoRA used 2,000 training examples, 200 validation examples, and 500 test examples for computational feasibility.

C. Metrics

Accuracy is easy to interpret but does not show whether errors are concentrated in particular classes. Precision, recall, and F1 were therefore calculated per class and summarized with weighted averaging. For class c,

$$F1_c = 2\frac{P_c R_c}{P_c + R_c}. \tag{5}$$

Weighted F1 averages $F1_c$ using the true class support as weights. Macro F1 gives every class equal weight. Weighted F1 was the main metric recorded throughout the notebooks, while macro F1 was reconstructed from the saved Llama prediction files to expose the effect of class imbalance in the QLoRA subset. Reporting both is important because a high weighted score can coexist with weak minority-class performance [22].

D. Comparability Rules

The project contains experiments with different evaluation sizes and, in one case, different input information. To avoid misleading rankings, the analysis follows three rules:

1) Results on 7,317 or 7,532 documents are presented as the main full-test comparison, with the minor filtering difference explicitly disclosed.
2) Prompted Llama-3 results on 2,000 balanced examples are compared with one another, rather than treated as exact full-test equivalents.
3) QLoRA on 500 examples is reported as an exploratory task-adaptation result, not as proof that it outperformed BERT.

The full-context Llama experiment includes headers, footers, and quotations. It is therefore used diagnostically to examine the value of metadata and thread context, rather than as a directly comparable competitor to the cleaned models.

IV. Experimental Workflow and Reproducibility

The project began locally in VS Code with data inspection, preprocessing, and an initial Word2Vec model. This was sufficient for classical methods, but the workflow became restrictive once BERT and Llama-3 were introduced. I moved to Google Colab for GPU access and later used Colab Pro to access stronger accelerators, including an A100 runtime for the largest experiments. This progression affected the methodology: it made repeated BERT runs, 4-bit Llama inference, and QLoRA feasible, but it also created practical limits on evaluation size and training duration.

The final work was separated into three notebooks: one for Tasks 1 and 2, one for Tasks 3 and 4, and one for the bonus QLoRA experiment. Weights & Biases (W&B) was added as the number of runs grew. It preserved configurations and metrics that would otherwise be scattered across notebook output. Random seeds and stratified splits were used where practical, and prediction-level outputs were saved for the Llama experiments. The code relies on gensim, PyTorch, scikit-learn, Hugging Face Transformers, bitsandbytes, PEFT, and TRL [23]–[25].

The workflow also revealed an important difference between reproducibility and repeatability. Fixed seeds and saved configurations improve repeatability, but a single run still does not estimate variance. GPU kernels, library versions, data order, and random initialization can produce small changes. The final claims therefore distinguish robust, large differences from small gains that require repeated-seed confirmation.

V. Methods

A. Shared Simple Neural Network

The classical experiments used a feed-forward neural network with two linear layers and a ReLU activation:

$$\mathbf{z} = W_2\,\mathrm{ReLU}(W_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2. \tag{6}$$

The output dimension was 20 and cross-entropy loss was used. Keeping this classifier simple served the experimental goal: differences should primarily reflect the input representation rather than a more complex architecture. Word2Vec document vectors used an input dimension of 100 and a 64-unit hidden layer. TF–IDF used 5,000 input features and a 128-unit hidden layer; the n-gram experiment used 10,000 features. Adam with a learning rate of 10^{-3} trained the models for ten epochs.

B. Word2Vec Epoch Sweep

Skip-gram Word2Vec models were trained on the cleaned training tokens with vector size 100, context window 5, minimum count 2, and one worker. The epoch values were 1, 3, 5, 10, and 20. Each embedding model was converted to mean-pooled document vectors using Eq. (3) and evaluated with the shared neural network.

Two forms of evidence were collected. First, downstream accuracy and weighted F1 measured whether the embedding space supported classification. Second, nearest neighbors for words such as *car*, *computer*, and *hockey* provided a qualitative view of local semantic organization. The assignment also requested a visual comparison of vector spaces. Initial PCA plots were generated, but independently trained spaces and independently fitted PCA projections cannot be interpreted as literal vector movement: embedding spaces can rotate or reflect while preserving relationships. A defensible coordinate-level comparison would align shared vocabulary with orthogonal Procrustes before using one joint projection [26]. Because trained checkpoints were not saved, the report does not manufacture an aligned plot after the fact. It uses the available neighborhood evolution and performance curves and treats the original PCA panels as qualitative only.

C. TF–IDF and N-Gram Experiment

The unigram baseline used 5,000 TF–IDF features and English stopword filtering. The extra experiment expanded the vocabulary to 10,000 features, used `ngram_range=(1,2)`, and applied sublinear term frequency. The classifier remained the same two-layer form. This directly answered whether a richer lexical representation could improve results without replacing the network.

D. BERT Classification and Hyperparameter Sweep

The supervised Transformer baseline used *bert-base-uncased* with a 20-class sequence-classification head. Documents were tokenized with padding and truncation. The main run used a maximum sequence length of 256, learning rate 2×10^{-5}, batch size 8, weight decay 0.01, and three epochs. Evaluation and logging occurred at epoch boundaries.

Four additional runs varied learning rate, epochs, and maximum length. Batch size remained 8. The sweep was intentionally small because each BERT run required substantially more compute than the classical models. It was designed to test plausible interactions: whether 256 tokens preserved useful evidence, whether 3×10^{-5} adapted faster, and whether a third epoch helped or overfit.

E. Task-Adaptive MLM Followed by Classification

For continued pretraining, the cleaned 20 Newsgroups training text was tokenized with BERT's tokenizer, concatenated, and grouped into blocks of 128 tokens. Fifteen percent of token positions were selected by the MLM data collator. The base model trained for two epochs with learning rate 5×10^{-5}, batch size 32, weight decay 0.01, and mixed precision when available. The adapted checkpoint was then loaded into a 20-class sequence-classification model and fine-tuned for three epochs with learning rate 2×10^{-5} and batch size 8.

This two-stage design separates unsupervised exposure to the task distribution from supervised label learning. The MLM loss measures token prediction rather than classification, so a lower MLM loss is not itself evidence of a better classifier. The relevant test is whether the adapted initialization improves accuracy and F1 after the same downstream fine-tuning process.

F. Llama-3 Zero-Shot, Few-Shot, and Full Context

The prompt experiments used *Meta-Llama-3-8B-Instruct* loaded in 4-bit NF4 quantization with double quantization and half-precision computation. Greedy generation was used with temperature 0 and four new tokens. The requested output was a single numerical label, and a parser validated that the first generated item was a class ID from 0 to 19. The zero-shot and few-shot runs both had a 0% unparsed rate, so their low scores reflect label choices rather than output-format failure.

The zero-shot prompt listed the 20 labels and asked for exactly one number. The few-shot prompt added one training example for each class. Each cleaned document was shortened to 1,200 characters to control context and runtime. The full-context diagnostic used the original posts, retained metadata and quotations, and allowed 2,000 characters.

G. QLoRA Fine-Tuning

The bonus experiment fine-tuned the same Llama-3 checkpoint on prompt–completion pairs. Documents were limited to 800 characters and the model generated label names. The quantized base model remained frozen while LoRA adapters targeted the attention projections (query, key, value, and output) and feed-forward projections (gate, up, and down). The adapter rank was 16, scaling factor 32, and dropout 0.05.

Training used one epoch, per-device batch size 1, gradient accumulation of 8, gradient checkpointing, a 2×10^{-4} learning rate, cosine scheduling, and completion-only loss. These choices were driven by memory and time constraints. They make the experiment a useful feasibility test, but not a fully tuned replacement for the main models.

Table II summarizes the preprocessing choices, training data, evaluation sets, and comparison conditions for each experimental family. Because several models were evaluated on different subsets or with different contextual information, these distinctions are important when interpreting the final results.

VI. Results

A. Main Full-Test Comparison

Table III and Fig. 1 summarize the primary results. The ranking is clear, but the gaps tell a more useful story than the order alone. Word2Vec improved as its embeddings trained longer. TF–IDF then exceeded the 20-epoch Word2Vec model, and bigrams improved the sparse representation further. BERT added approximately four weighted-F1 points over the

TABLE II
SUMMARY OF EXPERIMENTAL SETTINGS AND EVALUATION CONDITIONS.

Method	Representation / objective	Main settings	Test size	Comparability note
Word2Vec + NN	Mean of 100-d skip-gram word vectors	Window 5; min count 2; 1, 3, 5, 10, 20 embedding epochs; NN trained 10 epochs	7,532	Original full test; token cleaning can produce zero vectors for empty documents
TF–IDF + NN	Sparse unigram weights	5,000 features; English stopwords; 128 hidden units	7,532	Same original full-test size as main Word2Vec runs
TF–IDF (1,2)-grams + NN	Sparse unigram and bigram weights	10,000 features; sublinear TF; empty-document variant	7,532 / 7,317	Best classical result uses cleaned non-empty test documents
BERT classifier	Supervised contextual encoder	Max length 256; LR 2×10^{-5}; batch 8; 3 epochs	7,317	Main cleaned full-test reference
MLM-adapted BERT	Continued MLM, then supervised classification	MLM: 2 epochs, LR 5×10^{-5}; classifier: 3 epochs, LR 2×10^{-5}	7,317	Directly comparable with main BERT test set
Llama-3 prompting	Generative zero-/few-shot classification	4-bit; greedy decoding; 1,200 chars; balanced 100 examples/class	2,000	Balanced subset; not a full-test evaluation
Llama-3 full context	Zero-shot generation with raw post context	Headers, footers, quotes retained; 2,000 chars	2,000	Diagnostic only because input information differs
Llama-3 QLoRA	Quantized causal LM with low-rank adapters	Rank 16; 2,000 train; 1 epoch; 800 chars	500	Smaller, imbalanced subset; exploratory result

TABLE III
MAIN CLASSICAL AND BERT RESULTS.

Model	n	Acc.	Wt. F1
Word2Vec, 1 epoch + NN	7,532	26.43	0.2349
Word2Vec, 3 epochs + NN	7,532	45.30	0.4384
Word2Vec, 5 epochs + NN	7,532	53.05	0.5182
Word2Vec, 10 epochs + NN	7,532	59.76	0.5905
Word2Vec, 20 epochs + NN	7,532	62.61	0.6176
TF–IDF unigram + NN	7,532	64.23	0.6458
TF–IDF (1,2)-gram + NN	7,532	66.95	0.6720
TF–IDF (1,2), no empty docs	7,317	68.75	0.6870
BERT fine-tuned	7,317	73.28	0.7289
BERT MLM then fine-tuned	7,317	**74.01**	**0.7370**

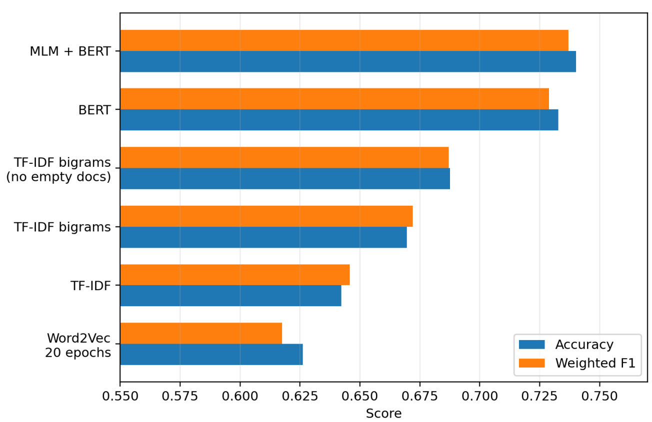

Fig. 1. Accuracy and weighted F1 for the principal classical and BERT experiments. Evaluation size is shown in Table III; the 7,317-document rows exclude cleaned empty documents.

strongest TF–IDF model. Continued MLM produced a smaller additional gain of 0.0081 weighted F1.

The strongest directly comparable model is therefore the

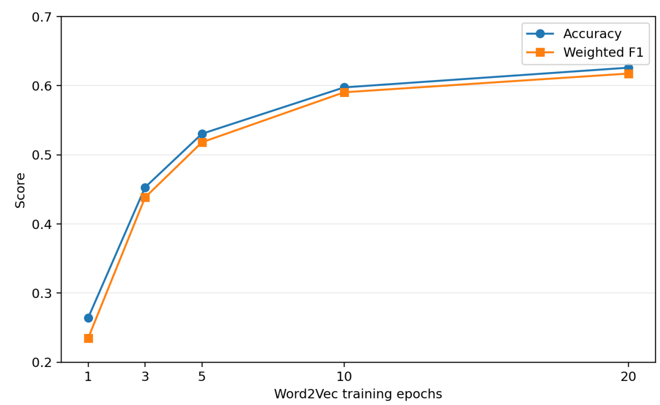

Fig. 2. Downstream simple-neural-network performance as Word2Vec embedding training increases.

MLM-adapted BERT classifier. Its improvement over standard BERT is meaningful as a direction, but small enough that it should not be called conclusive without repeated seeds or paired confidence intervals. The larger and more reliable findings are that contextual supervised BERT clearly exceeded the classical models, and that the strongest sparse baseline remained much closer to BERT than prompt-only Llama-3 did.

B. Word2Vec: Training Changed Both Utility and Neighborhoods

Figure 2 shows a steep improvement between one and five Word2Vec epochs, followed by diminishing returns. Accuracy rose from 26.43% after one epoch to 62.61% after twenty, while weighted F1 rose from 0.2349 to 0.6176. The model was not simply benefiting from a longer neural classifier training schedule; the embedding model itself was the controlled variable across the sweep.

TABLE IV
SELECTED WORD2VEC NEIGHBORHOOD EVOLUTION.

Word	Epoch	Selected nearest neighbors
car	1	still, side, least, little, bad
car	3	cars, tires, dealer, roads, motorcycle
car	5	tires, ford, cars, porsche, motorcycle
car	10	cars, porsche, volvo, dealer, tires
car	20	cars, sentra, audi, porsche, dealer
hockey	1	NHL, league, Russian, third, Sunday
hockey	20	NCAA, playoff, NHL, championship, tournament

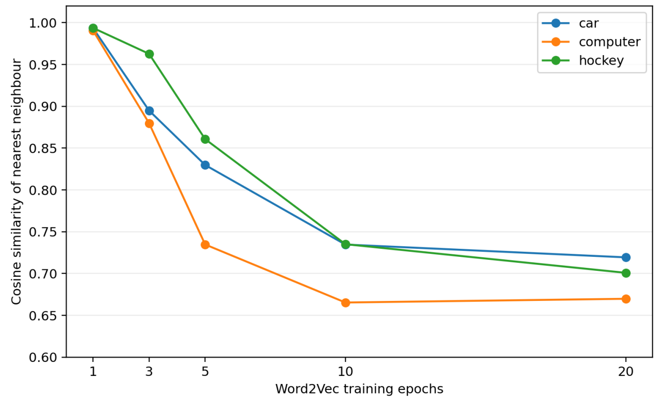

Fig. 3. Cosine similarity to the top nearest neighbor across Word2Vec training. Early near-one values accompanied diffuse, generic neighborhoods; later neighborhoods were more semantically specific.

The nearest-neighbor outputs explain why the one-epoch model was weak. For *car*, the highest-similarity neighbors included generic words such as *still*, *side*, *least*, *little*, and *bad*, all with similarities near 0.99. Such extreme and semantically diffuse similarities suggest an underdeveloped geometry in which many frequent words occupy nearly the same local region. By three and five epochs, automotive words such as *cars*, *tires*, *dealer*, *ford*, and *porsche* appeared. By twenty epochs, the neighborhood remained recognizably automotive, including *cars*, *sentra*, and *audi*.

The same pattern occurred for *hockey*. The later model associated it with *NCAA*, *playoff*, *NHL*, *championship*, and *tournament*. This is stronger evidence of semantic organization than a visually attractive but unaligned PCA plot. Table IV provides selected examples.

Figure 3 plots the similarity of each probe word to its single nearest neighbor. The decline from near-one similarities is not a loss of quality. In context, it indicates that the space became less collapsed and more selective: the best neighbor remained related, but unrelated words were no longer almost indistinguishable.

I expected this semantic improvement to make Word2Vec the strongest classical representation. It did not. Averaging helped the classifier recognize broad subject matter, but it smoothed away the exact lexical evidence needed to separate

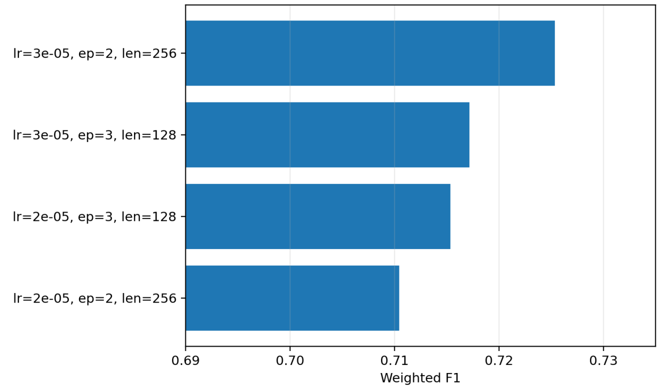

Fig. 4. Weighted F1 for the BERT hyperparameter experiments. Batch size was 8 in all four runs.

related newsgroups. This was the first point at which the results changed my interpretation of "semantic" features: richer word-level semantics do not guarantee a more discriminative document vector.

C. Why TF–IDF and Bigrams Were Strong

The unigram TF–IDF model reached 64.23% accuracy, already above the original 20-epoch Word2Vec result. Adding bigrams increased accuracy to 66.95%, and removing documents that became empty after cleaning raised it to 68.75%. This improvement occurred without replacing the two-layer network.

The result is consistent with the structure of the dataset. Many classes have technical vocabularies and recurring short expressions. A mean Word2Vec vector can make *IBM hardware*, *Mac hardware*, and *Windows* posts broadly similar because they share computer language. TF–IDF retains which words are present and how distinctive they are. Bigrams preserve useful local combinations. The model does not "understand" the post in a broad linguistic sense, but it receives features closely matched to the class boundaries.

This is also a reminder that classical baselines should not be treated as ceremonial. A simple, transparent representation can remain competitive when the task is lexically driven. More complex contextual models should justify their extra computational cost through matched improvements, not through their general reputation.

D. BERT and the Effect of Continued MLM

The supervised BERT model reached 73.28% accuracy and 0.7289 weighted F1. The MLM-adapted model increased these values to 74.01% and 0.7370. Figure 4 shows the additional hyperparameter runs. The best sweep configuration used learning rate 3×10^{-5}, two epochs, and maximum length 256, reaching 0.7253 weighted F1. It approached but did not exceed the main run.

The hyperparameter search explored the effects of learning rate, sequence length, and training duration. As shown in Table V, the strongest configuration within the four-run sweep

TABLE V
BERT HYPERPARAMETER EXPERIMENTS.

LR	Ep.	Len.	Acc.	Wt. F1
3×10^{-5}	2	256	72.87	0.7253
3×10^{-5}	3	128	72.05	0.7172
2×10^{-5}	3	128	71.83	0.7154
2×10^{-5}	2	256	71.87	0.7104

used a learning rate of 3×10^{-5}, a maximum sequence length of 256 tokens, and two training epochs, reaching a weighted F1 of 0.7253. This result approached but did not exceed the separately executed main BERT run, which used a learning rate of 2×10^{-5}, a maximum length of 256 tokens, and three epochs. The sweep therefore provided exploratory evidence about plausible settings rather than serving as a formal model-selection procedure for all subsequent experiments.

Longer sequences were particularly helpful in the two-epoch runs, which is plausible for discussion posts where important evidence may appear well beyond the opening paragraph. More epochs did not automatically improve performance. However, because multiple hyperparameters changed simultaneously and each setting was evaluated only once, the sweep should be interpreted as exploratory evidence rather than a precise estimate of individual hyperparameter effects.

The MLM stage used 9,912 training documents and achieved an evaluation loss of 2.3920 before classification fine-tuning. Its final gain over base BERT is small, which is also theoretically sensible. The base checkpoint was already pretrained on broad English, and 20 Newsgroups is not as distant from general text as clinical or legal corpora. The target corpus added forum-style language and specialist terminology, but the supervised classification objective already provided a strong task-specific learning signal. The main learning is not that continued pretraining guarantees a large gain. It is that it provided a plausible, low-magnitude improvement consistent with the limited distribution shift.

Figure 5 presents the confusion matrix for the final BERT classifier. Most errors occurred between semantically related categories, particularly within the sports, computing, and religion families. This suggests that the model learned the broader thematic structure of the corpus but occasionally struggled with fine-grained distinctions between closely related topics.

E. Prompted Llama-3: Valid Outputs, Weak Boundaries

Table VI separates the Llama experiments from the full-test ranking. Zero-shot prompting achieved 34.20% accuracy and 0.3147 macro/weighted F1 on the balanced subset. Few-shot prompting decreased accuracy to 31.75% and F1 to 0.2814. Full context improved zero-shot accuracy to 37.75% and F1 to 0.3434. Because the three prompted subsets contain exactly 100 examples per class, macro and weighted F1 are identical.

Figure 6 compares the performance of the Llama-3 variants. Prompt-only approaches remained substantially weaker than supervised BERT models, while QLoRA produced a large improvement, supporting the hypothesis that task-specific adaptation is more important than model scale alone.

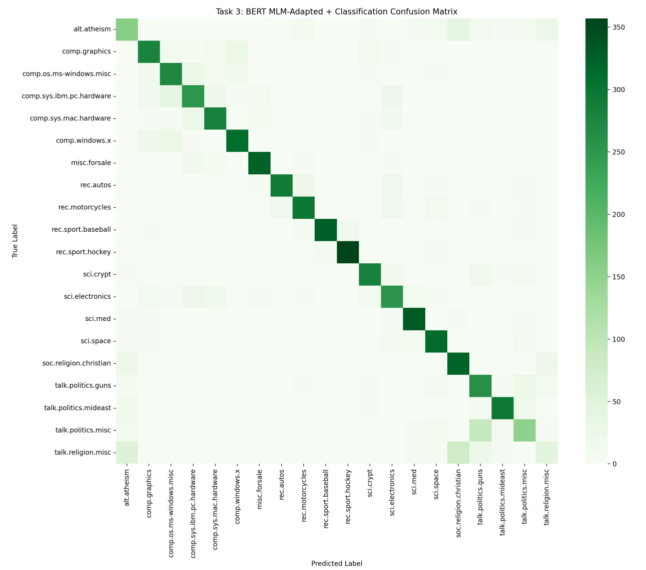

Fig. 5. Count confusion matrix for BERT after task-adaptive MLM and classification fine-tuning. Dense errors remain concentrated among related class families.

TABLE VI
LLAMA-3 PROMPT AND QLORA RESULTS.

Method	n	Acc.	Macro F1	Wt. F1
Zero-shot, cleaned	2,000	34.20	0.3147	0.3147
Few-shot, cleaned	2,000	31.75	0.2814	0.2814
Zero-shot, full context	2,000	37.75	0.3434	0.3434
QLoRA, cleaned subset	500	**75.00**	0.6894	**0.7434**

The parser did not fail for the prompt-only runs. Llama-3 consistently returned valid label numbers, yet it often selected the wrong one. This distinction matters. Prompt engineering solved the output-format problem, not the classification problem.

The confusion matrices in Figs. 7 and 8 show systematic broad-category bias. Zero-shot Llama-3 confused hockey with baseball 67 times, Middle East politics with gun politics 57 times, miscellaneous politics with gun politics 56 times, atheism with Christianity 54 times, and miscellaneous religion with Christianity 50 times. These are not random mistakes. The model often identified the semantic neighborhood but failed to reproduce the dataset's narrower decision boundary.

Few-shot performance was my most unexpected result. Adding examples seemed like the obvious way to clarify the label space. In practice, twenty demonstrations made the prompt long, while one example per category could not capture the diversity within each class. The examples may have acted as lexical anchors. For instance, a single representative of `comp.windows.x` cannot define its full boundary against graphics, Windows operating-system, and hardware posts. The result supports a cautious view of in-context learning:

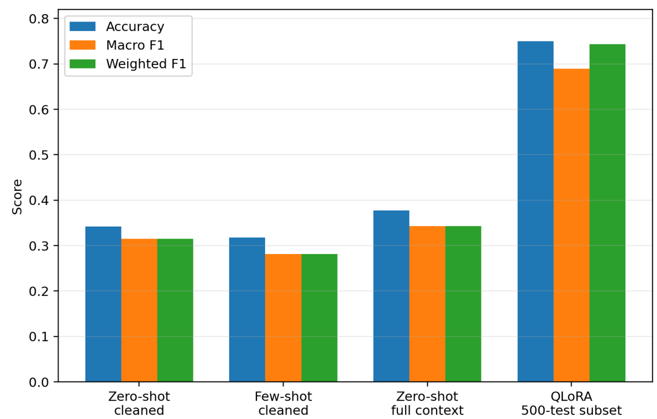

Fig. 6. Accuracy, macro F1, and weighted F1 for Llama-3 prompting and QLoRA. The QLoRA macro–weighted gap reflects unequal class support in its 500-document subset.

demonstrations change the model's behavior, but they do not necessarily teach a stable classifier.

Full context helped by supplying more evidence, but some of that evidence was metadata and quoted thread material deliberately removed elsewhere. The improvement therefore reveals that Llama-3 could exploit these cues. It does not prove that the model understood the cleaned post better. This is a useful distinction between better modeling and an easier information setting.

F. QLoRA: Stronger Alignment, Still an Unequal Comparison

QLoRA reached 75.00% accuracy and 0.7434 weighted F1, a sharp increase over prompt-only Llama-3. This supports the hypothesis that Llama-3 possessed useful language knowledge but was poorly aligned to the exact label mapping. Once trained on document–label pairs, it no longer had to infer the mapping afresh from a prompt.

The macro F1 of 0.6894 is substantially lower than the weighted F1 because the 500-document subset was not perfectly balanced. The adapter performed better on classes with more support and more recognizable patterns. The remaining confusion pairs were still concentrated among related categories: miscellaneous politics was predicted as gun politics ten times, Mac hardware as IBM PC hardware seven times, and several religion and computer pairs remained difficult.

The scientifically defensible conclusion is not that QLoRA beat BERT. Its evaluation used 500 documents rather than 7,317, its class supports differed, and its training subset was much smaller and separately sampled. It shows that parameter-efficient task adaptation can transform Llama-3's behavior and deserves a matched evaluation, which is a meaningful result. To better understand the limitations of prompt-only classification, Table VII lists the most frequent Llama-3 confusion pairs. Many errors occurred between classes sharing broad thematic content, indicating that the model often recognized the general subject area but failed to identify the precise newsgroup category.

VII. Discussion

A. RQ1: Representation Quality Is Task-Dependent

The classical experiments answer the first research question directly. More Word2Vec training improved the embedding neighborhoods and the neural classifier. The effect was large enough to demonstrate real representation learning: 20-epoch embeddings were not merely numerically different from one-epoch embeddings; they grouped automotive and sports vocabulary more coherently and produced much better downstream predictions.

Nevertheless, TF–IDF was stronger. This result is not a contradiction. Word2Vec answers a semantic similarity question: which words appear in comparable contexts? TF–IDF answers a discrimination question: which terms are unusually informative in this document relative to the corpus? The second question is closely aligned with a topic benchmark whose classes have distinctive vocabulary. Mean pooling further disadvantaged Word2Vec by removing word order and term importance. The n-gram result reinforces this explanation because adding short phrases improved the sparse model without adding contextual inference.

The experiment changed my initial assumption that dense vectors were inherently more advanced and therefore likely to win. Dense representations are compact and semantically expressive, but an expressive feature can still be poorly aggregated for a document-level decision. The comparison taught me to ask what information reaches the classifier, not only how sophisticated the representation sounds.

B. RQ2: Continued MLM Helped, but Modestly

The MLM-adapted BERT model improved weighted F1 from 0.7289 to 0.7370. This supports the direction of task-adaptive pretraining, but the gain is small relative to the jump from classical models to BERT. There are several plausible reasons. BERT was already pretrained on large general-English corpora. The target corpus contains technical and historical forum language, but it is not a completely new linguistic domain. Supervised fine-tuning on nearly ten thousand documents also supplies substantial task signal.

The result illustrates a useful distinction between pretraining and fine-tuning. MLM teaches the model to predict language within the target distribution; it does not directly teach the 20 decision boundaries. Classification fine-tuning still performs that alignment. Continued pretraining improves the starting point, not the objective itself. This is why a small gain can still be coherent with the theory.

At the same time, a one-run difference of 0.0081 could partly reflect random variation. The strongest wording is therefore: the experiment provides evidence consistent with a benefit from task-adaptive MLM, but repeated seeds and paired analysis are needed before treating the exact gain as stable.

C. RQ3: Objective Alignment Was More Important Than Scale

Prompt-only Llama-3 was far larger than BERT but much weaker on this closed-set task. The comparison demonstrates

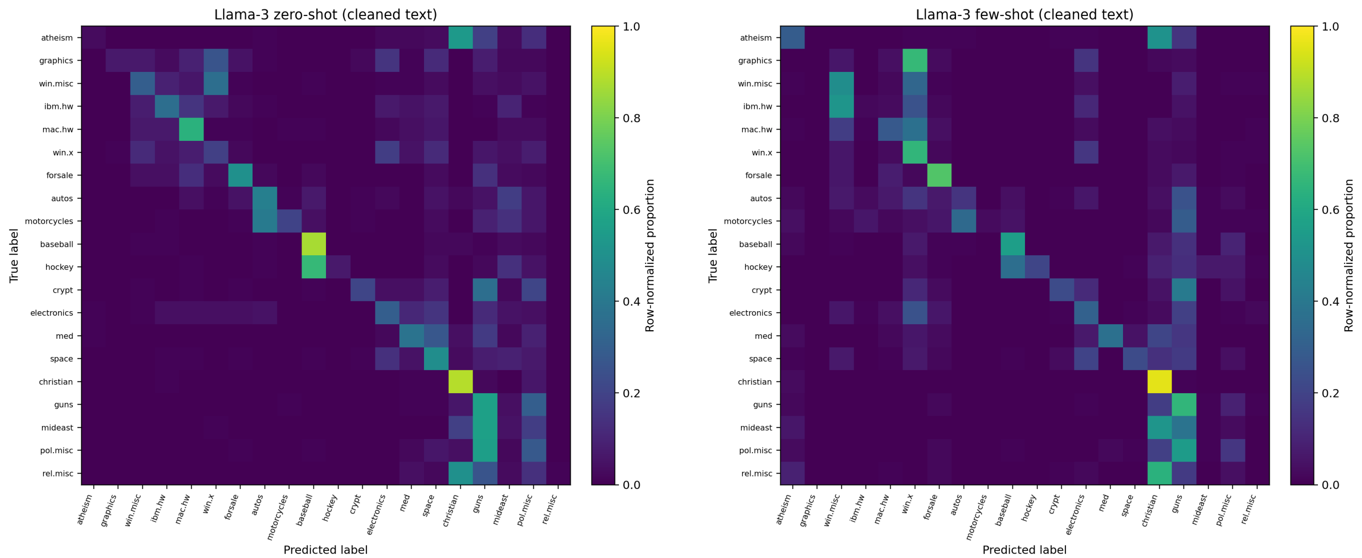

Fig. 7. Row-normalized confusion matrices for cleaned-text Llama-3 prompting. Left: zero-shot. Right: one-example-per-class few-shot. Few-shot demonstrations increased concentration in several broad labels rather than sharpening all class boundaries.

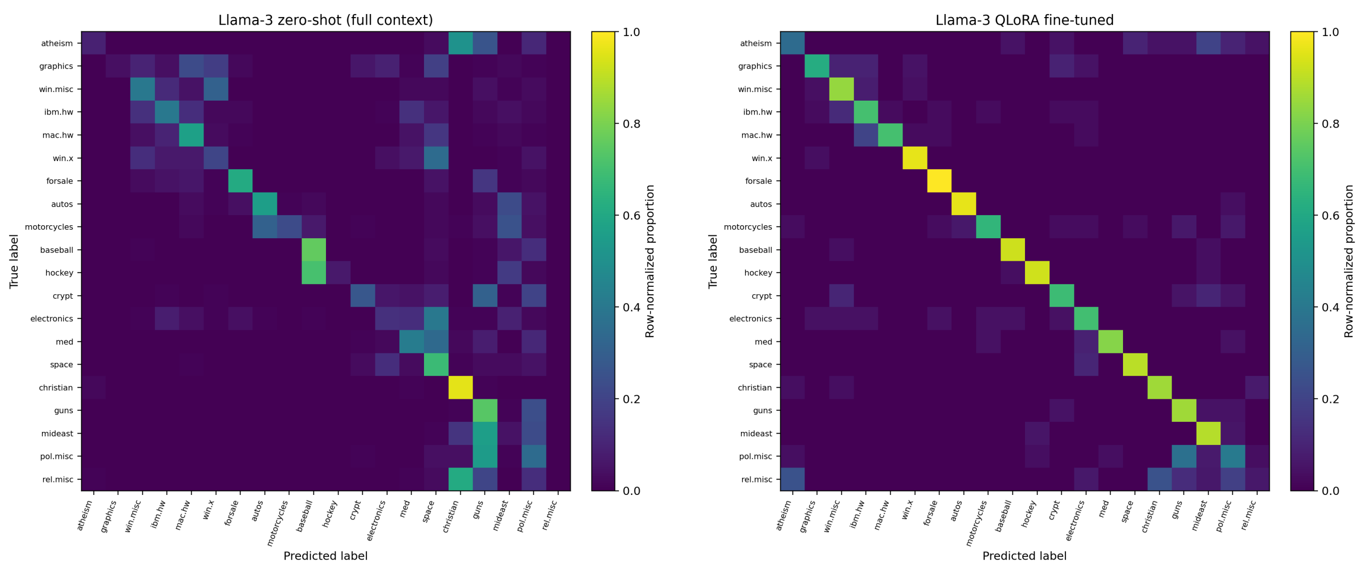

Fig. 8. Row-normalized diagnostic matrices. Left: zero-shot Llama-3 with headers, footers, quotations, and longer input. Right: QLoRA fine-tuning on the smaller cleaned subset. QLoRA produces a much stronger diagonal, while the full-context prompt retains broad confusion patterns.

why parameter count alone is a poor explanation. BERT used a classification head and gradient updates on thousands of labeled examples. Llama-3 generated a short string after reading an instruction. It had to infer the dataset taxonomy, respect the output format, and distinguish categories that are close even for a supervised model.

The zero-shot confusion patterns suggest that Llama-3 had broad semantic competence. It often chose a neighboring class rather than an unrelated one. Yet the benchmark rewards exact boundaries. A model that recognizes "this is about religion" but chooses Christianity instead of atheism receives no partial credit. BERT's advantage was not simply that it represented language contextually; it was trained to make exactly this distinction.

QLoRA supports this interpretation. Lightweight supervised adaptation changed Llama-3 from a weak prompt classifier to a strong subset classifier. The model did not become larger. The task connection became stronger. This is the clearest evidence in the project that task alignment mattered more than model scale alone.

D. Negative and Diagnostic Results Were Valuable

Few-shot prompting did not work as expected. Full context improved the score but changed the task. QLoRA produced the highest subset number but not a matched comparison. None of

TABLE VII
MOST FREQUENT OFF-DIAGONAL LLAMA-3 CONFUSIONS.

Experiment	True class	Predicted class	Count
Zero-shot cleaned	rec.sport.hockey	rec.sport.baseball	67
Zero-shot cleaned	talk.politics.mideast	talk.politics.guns	57
Zero-shot cleaned	talk.politics.misc	talk.politics.guns	56
Zero-shot cleaned	alt.atheism	soc.religion.christian	54
Few-shot cleaned	comp.graphics	comp.windows.x	67
Few-shot cleaned	talk.religion.misc	soc.religion.christian	64
Few-shot cleaned	comp.sys.ibm.pc.hardware	comp.os.ms-windows.misc	52
Full context	rec.sport.hockey	rec.sport.baseball	71
Full context	talk.religion.misc	soc.religion.christian	61
QLoRA	talk.politics.misc	talk.politics.guns	10
QLoRA	comp.sys.mac.hardware	comp.sys.ibm.pc.hardware	7

these findings should be hidden. They clarify what each result can and cannot support.

The few-shot run shows that examples can add ambiguity. The full-context run shows that metadata is predictive and that preprocessing defines the meaning of "classification performance." The QLoRA caveat shows why sample size and split construction belong beside every score. Together, these experiments moved the project beyond a leaderboard and toward a scientific explanation of behavior.

E. Efficiency and Practical Model Choice

Performance is not the only consideration. TF–IDF and a small neural network are inexpensive to train, easy to inspect, and strong enough to be credible. BERT requires more compute but delivers the best matched full-test performance. Llama-3 prompting requires substantial memory and slow autoregressive generation while producing low accuracy in this setup. QLoRA reduces the memory cost of fine-tuning, but its training and generation pipeline remains more complex than BERT classification.

For a practical 20 Newsgroups-like application, the evidence favors a staged decision. A TF–IDF baseline should be established first. A supervised encoder is justified when the additional accuracy, contextual robustness, or transfer value matters. A large generative model becomes more defensible when the application also requires explanation, flexible labels, or generation—or when parameter-efficient fine-tuning is evaluated fairly and delivers a clear benefit.

VIII. THREATS TO VALIDITY AND LIMITATIONS

A. Unequal Evaluation Sets

The principal limitation is that not all methods were evaluated on the same documents. Classical full-test runs used 7,532 examples, cleaned non-empty BERT runs used 7,317, prompted Llama-3 used a balanced 2,000-example subset, and QLoRA used 500 examples. The final report avoids one combined leaderboard, but unequal samples still limit cross-family conclusions. A matched test set is necessary to estimate the true BERT–QLoRA difference.

B. Single Runs and Statistical Uncertainty

Most training configurations were run once. Neural models vary with random initialization, batch order, data sampling, mixed precision, and hardware. Repeating the principal BERT and Word2Vec settings across at least three seeds would permit means and standard deviations. Because base BERT and MLM-adapted BERT predict the same test documents, paired bootstrap confidence intervals or an appropriate paired significance test could evaluate their small difference [27].

C. Effect of Sequence Truncation

Sequence length was one of the hyperparameters explored during BERT fine-tuning. Analysis of the cleaned test set revealed that 48.83% of documents contained more than 128 tokens, while 23.90% contained more than 256 tokens after tokenization. This indicates that truncation affected a large fraction of the dataset, particularly in the 128-token setting. The longer 256-token configuration therefore allowed BERT to process a greater proportion of each document and may explain the modest performance improvements observed in experiments using longer input lengths. Nevertheless, nearly a quarter of the evaluated documents remained truncated even at 256 tokens, suggesting that further gains could potentially be achieved with longer-context models.

D. Preprocessing Changes the Task

Removing headers, footers, and quotations discourages shortcut learning, but it also removes legitimate context. The full-context result demonstrates that the choice affects Llama-3. Token-level cleaning creates a second variation for classical models. These differences were motivated and documented, yet a fully controlled study would run every feasible model on both raw and cleaned settings.

E. Embedding Visualization

A two-dimensional projection can be persuasive but misleading. Independently trained Word2Vec spaces are not directly aligned, and PCA axes fitted separately can rotate. The qualitative conclusion that neighborhoods became more coherent is supported by nearest-neighbor lists and downstream performance. A future run should save checkpoints, align shared vocabulary using orthogonal Procrustes, and project the aligned spaces jointly before drawing movement arrows [26].

F. Metric Selection and Class Imbalance

Weighted F1 was appropriate for the main multi-class summaries, but it can hide poor performance on classes with low support. The reconstructed QLoRA macro F1 is lower than its weighted F1, demonstrating this risk. Future reporting should include macro F1, per-class F1, and confidence intervals for all final models.

G. Baseline Scope

The two-layer network was required by the assignment and intentionally fixed. It is not necessarily the strongest classifier for sparse text. Linear SVMs and logistic regression are established high-dimensional text baselines [28]. Including them would help separate representation quality from the suitability of the neural classifier. Similarly, weighted pooling such as SIF would provide a stronger static-embedding baseline [6].

IX. Future Work

The highest-priority extension is a matched evaluation of base BERT, MLM-adapted BERT, and QLoRA on the same cleaned 7,317 documents. The QLoRA run should also be repeated over several training seeds and subset samples. This would determine whether its strong result generalizes beyond one 500-document evaluation.

Prompt experiments should vary one design choice at a time. Useful ablations include label names versus numerical IDs, concise class descriptions, two or more examples per class, semantically retrieved demonstrations, random versus prototypical examples, and several demonstration orders. Calibration methods could test whether a few over-predicted classes are partly caused by prompt-induced label bias [15].

Error analysis should be organized by class family. For example, a hierarchical classifier could first choose among computer, recreation, science, politics, religion, and marketplace families, then choose a fine-grained label. This would match the observed error structure and reveal whether the models struggle more with broad recognition or within-family discrimination.

The classical side can also be strengthened. A linear SVM, character n-grams, and a weighted Word2Vec document representation would create more competitive low-cost baselines. On the Transformer side, RoBERTa could test whether the result reflects the general advantage of supervised encoders rather than BERT specifically [29].

Finally, efficiency should be measured explicitly. Training time, peak GPU memory, inference throughput, and energy use would turn the comparison into a practical model-selection study. A one-point gain may be valuable in one context and unjustifiable in another.

X. Personal Learning and Methodological Reflection

The most important learning did not come from the highest score. It came from moments when the results contradicted an expectation.

First, I assumed Word2Vec would beat TF–IDF because it learned semantic relationships rather than simple frequencies. The epoch sweep partly supported that intuition: the embedding neighborhoods became visibly more meaningful and the classifier improved dramatically. However, TF–IDF still won. I learned that a representation can be semantically richer at word level but less useful after a lossy document-aggregation step. I also learned to be cautious with visualizations. Two PCA plots can look as if words “moved,” but without alignment they do not share a coordinate system. The nearest-neighbor lists and downstream curve were more trustworthy than a convenient picture.

Second, I expected few-shot prompting to improve Llama-3. Instead, it performed worse. That result forced me to inspect the prompt as part of the experimental system rather than treating the model as the only variable. Twenty examples made the prompt longer, one example per class was not representative, and the model could anchor on surface wording. Prompt engineering became less like writing a clever instruction and more like designing a sensitive input distribution.

Third, the similarity between strong TF–IDF and BERT results was initially surprising. BERT was better, but not by an overwhelming margin. The comparison made me appreciate why simple baselines are scientifically important. They show how much of a benchmark can be solved by lexical evidence and prevent a more complex model from receiving credit for difficulty that was never present.

Fourth, continued MLM and QLoRA taught me two different meanings of adaptation. BERT already had an objective close to classification and a strong pretrained encoder, so continued MLM produced a small gain. Prompted Llama-3 had no learned connection to the dataset labels, so supervised adapters changed the result much more. Adaptation is not one uniform technique; its impact depends on the gap between the starting model and the task.

The project also improved my practical research process. Moving from local execution to Colab and then to an A100 made larger experiments possible, but it did not remove the need to plan evaluation size and preserve outputs. W&B tracking and saved CSV predictions became essential once the number of runs grew. In hindsight, I would save every trained Word2Vec checkpoint, prediction array, token-length statistic, and random seed from the beginning. Scientific reporting is much easier when reproducibility is designed into the experiment rather than reconstructed at the end.

XI. Conclusion

This study compared sparse lexical features, static embeddings, contextual encoders, prompted large language models, and parameter-efficient fine-tuning on one fine-grained classification task. The best matched full-test result came from BERT after task-adaptive MLM and supervised fine-tuning, with 74.01% accuracy and 0.7370 weighted F1. TF–IDF bigrams reached 68.75% accuracy and remained a strong, efficient baseline. Word2Vec improved substantially with training but was limited by unweighted document averaging.

Prompt-only Llama-3 produced valid outputs but weak exact-label decisions. Few-shot demonstrations did not automatically help, and retaining full context improved performance partly by adding metadata unavailable to the cleaned models. QLoRA then changed the result sharply on a smaller subset, showing that the model's general knowledge became much more useful after direct task adaptation.

The central conclusion is therefore not that one representation or architecture is universally best. The evidence supports a more specific claim: model performance depended on whether the representation preserved the cues that separated the classes and whether the training objective explicitly aligned the model to those boundaries. Ultimately, the findings offer a clear cautionary tale for modern NLP pipelines: when it comes to navigating the narrow, high-entropy decision boundaries of fine-grained document classification, model scale alone fails where targeted task alignment succeeds. For practitioners, the results suggest that strong lexical baselines and supervised encoder models should remain the first point of comparison before adopting larger generative models for fine-grained document classification tasks.

Reproducibility, Tools, and AI Disclosure

The full repository, notebooks, outputs, and report sources are available at `github.com/Lamiorkor/NALAPRO_Final_Project`. Experiment tracking is available through the W&B project linked in the repository. The implementation used Python, Google Colab, NumPy, pandas, scikit-learn, NLTK, gensim, PyTorch, Hugging Face Transformers and Datasets, bitsandbytes, PEFT, TRL, Matplotlib, and Weights & Biases.

AI assistance was used for debugging, explaining errors and concepts, organizing experiments, and supporting report editing. The assistance did not replace the executed experiments. I reviewed and adapted the code, checked reported values against saved outputs, and retained responsibility for the methodology, interpretations, and final text.

Appendix A
Label Families Used in Error Analysis

The 20 labels were grouped only for interpretation; the models still predicted the original 20 classes. The groups were:

- **Computer:** `comp.graphics`, `comp.os.ms-windows.misc`, `comp.sys.ibm.pc.hardware`, `comp.sys.mac.hardware`, `comp.windows.x`.
- **Recreation and marketplace:** `rec.autos`, `rec.motorcycles`, `rec.sport.baseball`, `rec.sport.hockey`, and `misc.forsale`.
- **Science:** `sci.crypt`, `sci.electronics`, `sci.med`, and `sci.space`.
- **Politics:** `talk.politics.guns`, `talk.politics.mideast`, and `talk.politics.misc`.
- **Religion:** `alt.atheism`, `soc.religion.christian`, and `talk.religion.misc`.

This grouping explains why several high-frequency errors are semantically close even though they are fully incorrect under exact-match evaluation.

Appendix B
Prompt Designs

The zero-shot prompt defined the task as classification of a 20 Newsgroups document, enumerated the numerical IDs and label names, inserted the shortened document, and instructed the model to return one label number without explanation. The system message described the model as a precise text-classification assistant.

The few-shot prompt used the same requested numerical output and added one labeled training document per class. Keeping the demonstrations and requested output in the same numerical format eliminated parsing inconsistencies observed during development. It did not, however, improve classification accuracy.

The full-context prompt warned that the input could contain headers, quotations, e-mail metadata, and thread context and instructed the model to use all available information. The QLoRA prompt instead listed label names and trained the completion to contain the correct label name. These differences are documented because prompt format is part of the model specification, not merely presentation text.

Appendix C
Key Hyperparameters

Table VIII summarizes the main hyperparameters used for the final reported configurations.

Acknowledgment

The author thanks God for the strength and ideas to complete this project, and Dr. Forooz Shahbazi Avarvand, her NALAPRO lecturer at HSLU, for the course structure and project guidance. Google Colab and Colab Pro provided the compute used for the Transformer and Llama-3 experiments.

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TABLE VIII
DETAILED HYPERPARAMETERS FOR THE FINAL REPORTED CONFIGURATIONS.

Experiment	Configuration
Word2Vec	Skip-gram; vector size 100; window 5; minimum count 2; workers 1; embedding epochs 1, 3, 5, 10, and 20. Mean pooling; NN hidden size 64; Adam LR 10^{-3}; batch 32; 10 NN epochs.
TF–IDF	Unigram: 5,000 features, English stopwords. N-gram: 10,000 features, range (1,2), sublinear TF. NN hidden size 128; Adam LR 10^{-3}; batch 64; 10 epochs.
BERT classifier	*bert-base-uncased*; max length 256; LR 2×10^{-5}; batch 8; 3 epochs; weight decay 0.01; epoch evaluation and logging.
BERT sweep	LR in $\{2, 3\} \times 10^{-5}$; epochs in {2,3}; max length in {128,256}; batch 8; best checkpoint selected by validation F1.
MLM adaptation	*bert-base-uncased*; tokenization length 256; grouped block size 128; MLM probability 0.15; LR 5×10^{-5}; batch 32; 2 epochs; weight decay 0.01. Downstream classifier: LR 2×10^{-5}; batch 8; 3 epochs.
Llama prompting	*Meta-Llama-3-8B-Instruct*; 4-bit NF4; double quantization; greedy decoding; temperature 0; 4 new tokens; 1,200 cleaned characters or 2,000 full-context characters.
QLoRA	Rank 16; alpha 32; dropout 0.05; target attention and MLP projection modules; 4-bit NF4 base; LR 2×10^{-4}; one epoch; batch 1; gradient accumulation 8; gradient checkpointing; cosine schedule; maximum length 768; completion-only loss.

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